TIØ4145 Corporate Finance Chapters 6 and 8

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Bond Valuation

- Chapter 6 learning objectives
  - You know **essential expressions** related to bonds.
  - You can **identify the cash flows** for both coupon bonds and zero-coupon bonds, and calculate the value for each type of bond.
  - You can calculate the **yield to maturity** for both coupon and zero-coupon bonds, and interpret its meaning for each.
  - Given coupon rate and yield to maturity, you can determine whether a coupon bond will sell at a **premium or a discount**. You can describe **the time path the bond's price will follow** as it approaches maturity.
  - You can illustrate the change in bond prices when interest rates change.
  - You can discuss **the sensitivity** of a bond prices **to** changes in **interest rates**.
  - You know the concept of **duration**.

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Bond Valuation

- Chapter 6 learning objectives cont.
  - You can calculate the price of *a* **coupon bond from a series of zero-coupon** bonds using the Law of One Price.
  - You can discuss the relation **between a corporate bond's expected return** and the yield to maturity; define **default risk** and explain how these rates incorporate default risk.
  - You can assess the creditworthiness of a corporate bond using its **bond rating**.

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Bond Terminology

Bond is a security sold by governments and corporations to raise money today in exchange for future repayments.

- **Bond Certificate**: States the terms of the bond
- **Maturity Date:** Final repayment date
- **Term:** The time remaining until the repayment date
- **Coupon:** Promised interest payments
- **Principal/Face Value:** Notional amount the bond issuer promises to repay at maturity and which is used to compute the interest payments
- **Coupon Rate:** Determines the amount of each coupon

Zero-Coupon Bonds

- Let's assume for now that bonds are **risk free**
- A **Zero-Coupon** is a bond that not make coupon payments.
- Always sells at a **discount** (a price lower than face value), so they are also called **pure discount bonds**
- **Treasury Bills** are U.S. government zero-coupon bonds with a maturity of up to one year.
- Suppose that a one-year, risk-free, zero-coupon bond with a $100,000 face value has an initial price of $96,618.36. The cash flows would be:

- Although the bond pays no "interest", your compensation is the difference between the initial price and the face value.

Yield to Maturity

- The discount rate that sets the present value of the promised bond payments equal to the current market price of the bond.

- Price of a Zero-Coupon bond: $$P = \frac{FV}{(1 + YTM_n)^n}$$

- Yield to Maturity of a Zero-Coupon Bond $$YTM_n = \left(\frac{FV}{P}\right)^{1/n} - 1$$

- Example: Suppose that the following zero-coupon bonds are selling at the prices shown below per $100 face value. Determine the corresponding yield to maturity for each bond.

Maturity	1 year	2 years	3 years	4 years
Price	$98.04	$95.18	$91.51	$87.14
YTM	2%	2.5%	3%	3.5%

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Question?

- What is the difference between:
  - Yeild to maturity
  - Discount rate
  - Interest rate
  - Policy rate

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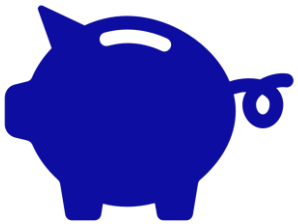
How it is all connected

Central bank sets the **policy rate**



Banks adjust their **interest rate**s (their cost of funds tied to the policy rate)

Investors constantly compare returns affecting **bond yields**

Anyone who needs to discount a stream of cashflows can use the return on the instrument with the same timing and risk as the **discount rate.**

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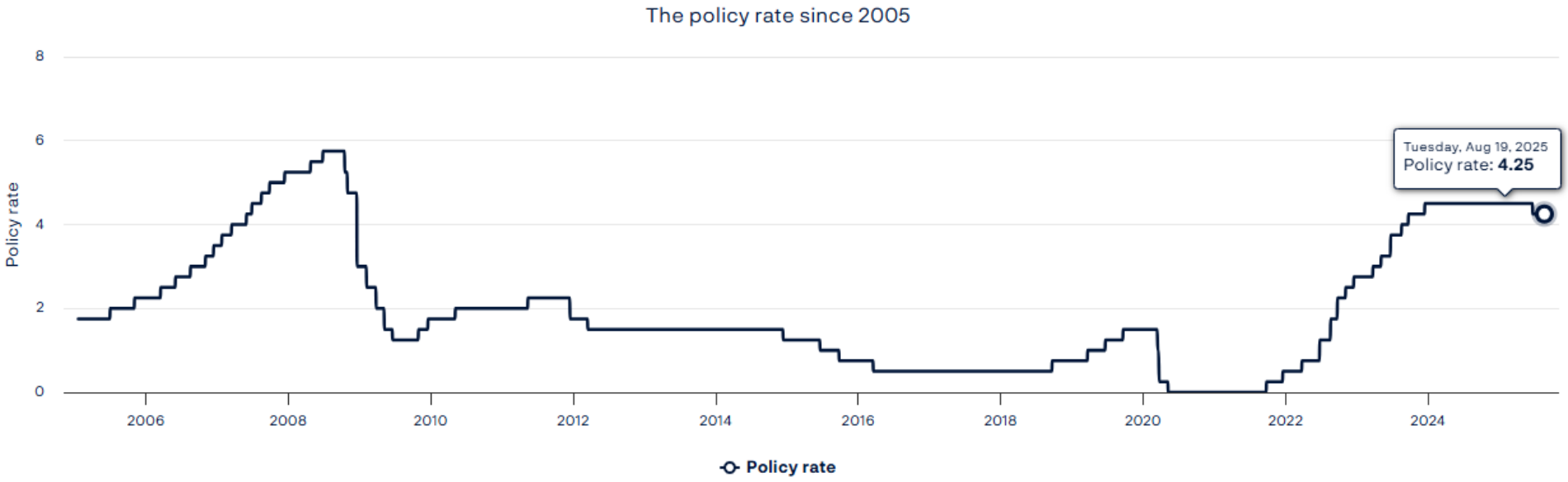
Policy rate in Norway



- *Source: Norges bank*

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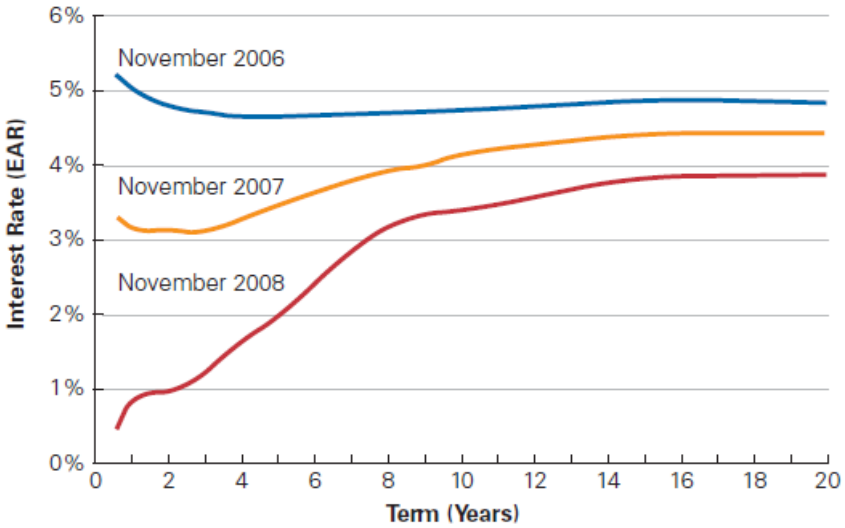
Yield curve recap

FIGURE 5.2 Term Structure of Risk-Free U.S. Interest Rates, November 2006, 2007, and 2008

Term (years)	Date Nov-06	Nov-07	Nov-08
0.5	5.23%	3.32%	0.47%
1	4.99%	3.16%	0.91%
2	4.80%	3.16%	0.98%
3	4.72%	3.12%	1.26%
4	4.63%	3.34%	1.69%
5	4.64%	3.48%	2.01%
6	4.65%	3.63%	2.49%
7	4.66%	3.79%	2.90%
8	4.69%	3.96%	3.21%
9	4.70%	4.00%	3.38%
10	4.73%	4.18%	3.41%
15	4.89%	4.44%	3.86%
20	4.87%	4.45%	3.87%



The figure shows the interest rate available from investing in risk-free U.S. Treasury securities with different investment terms. In each case, the interest rates differ depending on the horizon. (Data from U.S. Treasury STRIPS.)

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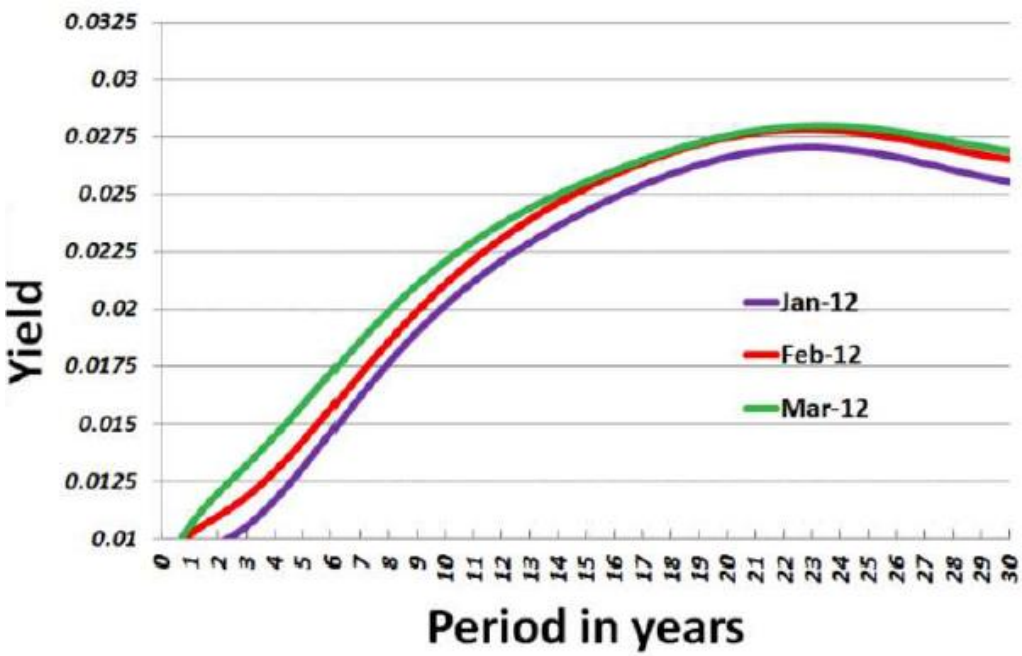
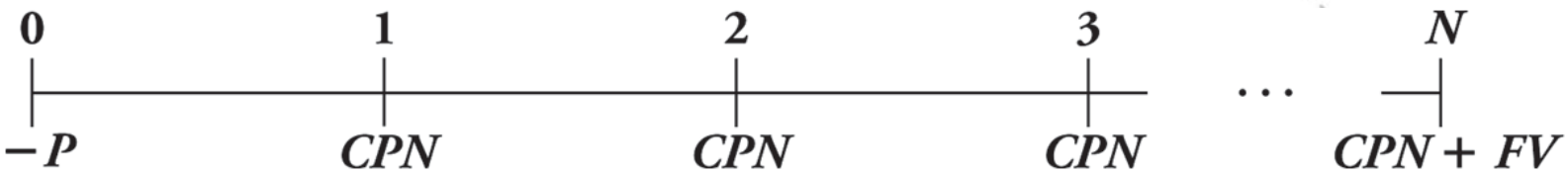
Risk-Free Interest Rates

- A default-free zero-coupon bond that matures on date *n* provides a risk-free return over the same period.
- Thus, the Law of One Price guarantees that the risk-free interest rate equals the yield to maturity on such a bond.
- **Spot Interest Rate:** Another term for a default-free, zero-coupon yield
- **Zero-Coupon Yield Curve:** A plot of the yield of risk-free zero-coupon bonds as a function of the bond's maturity date.



Coupon Bonds

- Pay **face value** at maturity and **regular coupon** interest payments



- **Treasury Notes**: U.S. Treasury coupon security with original maturities of 1–10 years
- **Treasury Bonds**: U.S. Treasury coupon security with original maturities over 10 years
- **Yield to Maturity**: The YTM is the *single* discount rate that equates the present value of the bond's remaining cash flows to its current price:

$$P = CPN \times \frac{1}{y}\left(1 - \frac{1}{(1+y)^N}\right) + \frac{FV}{(1+y)^N}$$

Example

- Consider the following semi-annual bond (Also study text book example 6.3):
  - $1000 face value
  - 7 years until maturity
  - 9 % coupon
  - Price is $1,080.55
- What is the bond's yield to maturity?

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Example

- Given the yield we can compute the price by the formula

$$P = CPN \times \frac{1}{y}\left(1 - \frac{1}{(1+y)^N}\right) + \frac{FV}{(1+y)^N}$$

- N = 7 years × 2 = 14
- CPN = (9% × $1000) ÷ 2 = $45
- P = $1,080.55

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Dynamic Behavior of Bond Prices

- **Discount:** A bond is selling at a discount if the price is less than the face value.
- **Par:** A bond is selling at par if the price is equal to the face value.
- **Premium:** A bond is selling at a premium if the price is greater than the face value.

- If a coupon bond trades at a discount, an investor will earn a return both from receiving the **coupons** and from receiving a **face value** that **exceeds** the **price** paid for the bond.
- If a bond trades **at a discount**, its **yield** to maturity will **exceed** its **coupon rate**.
- If a coupon bond trades at a premium it will earn a return from receiving the coupons but this **return** will be **diminished** by receiving a **face value less than the price** paid for the bond. Thus, its **yield** to maturity is **less** than its **coupon** rate.
- A bond trades **at par** when its coupon rate is **equal to** its yield to maturity.

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Example

Determining the Discount or Premium of a Coupon Bond

Problem

Consider three 30-year bonds with annual coupon payments. One bond has a 10% coupon rate, one has a 5% coupon rate, and one has a 3% coupon rate. If the yield to maturity of each bond is 5%, what is the price of each bond per \$100 face value? Which bond trades at a premium, which trades at a discount, and which trades at par?

Solution

We can compute the price of each bond using Eq. 6.5. Therefore, the bond prices are

$$P(10\%\text{ coupon}) = 10 \times \frac{1}{0.05}\left(1 - \frac{1}{1.05^{30}}\right) + \frac{100}{1.05^{30}} = \$176.86 \quad \text{(trades at a premium)}$$

$$P(5\%\text{ coupon}) = 5 \times \frac{1}{0.05}\left(1 - \frac{1}{1.05^{30}}\right) + \frac{100}{1.05^{30}} = \$100.00 \quad \text{(trades at par)}$$

$$P(3\%\text{ coupon}) = 3 \times \frac{1}{0.05}\left(1 - \frac{1}{1.05^{30}}\right) + \frac{100}{1.05^{30}} = \$69.26 \quad \text{(trades at a discount)}$$

Changes in bond prices

- After the issue date, the market price of a bond generally changes over time for two reasons:
  - **Pull-to-par effect:** Holding fixed the bond's yield to maturity, the present value of the bond's remaining cash flows changes as the time to maturity decreases.
  - **Yield effect:** Changes in market interest rates affect the bond's yield to maturity and its price.

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Time and Bond Prices

- Holding all other things constant, a **bond's yield to maturity** will **not change over time**.
- Holding all other things constant, the price of **discount or premium** bond will **move towards par** value over time.
- If a bond's yield to maturity has not changed, then the **IRR** of an investment in the bond **equals** its **yield to maturity** even if you sell the bond **early**.
- Study textbook Example 6.6

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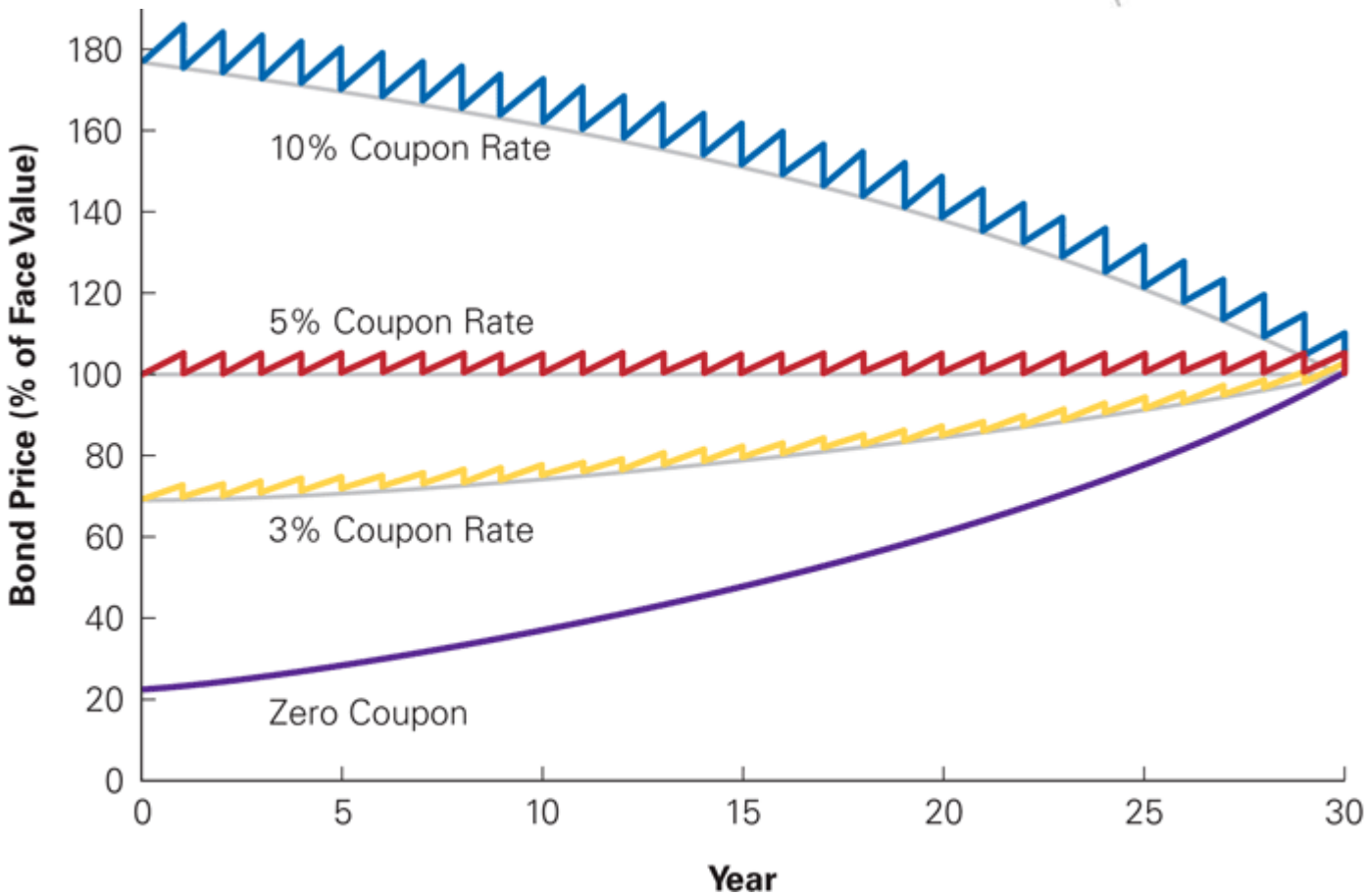
Time and Bond Prices



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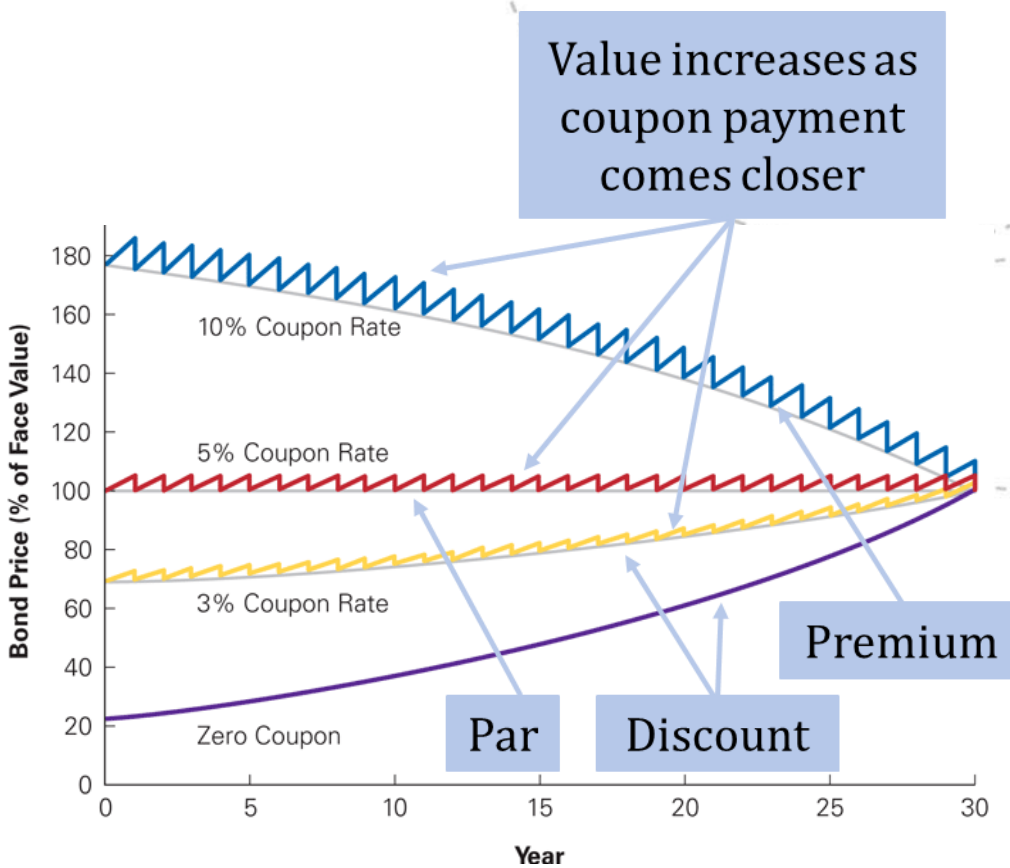
Time paths for bond prices

Traders and marketplaces often quote **clean price** i.e. remove sawtooth effects from **dirty price**:

$$Clean\ price\ =\ Cash\ (dirty)\ price - Accrued\ interest$$

$$Accrued\ interest = Coupon\ amount \times \left(\frac{Days\ since\ last\ coupon\ payment}{Days\ in\ current\ coupon\ period}\right)$$



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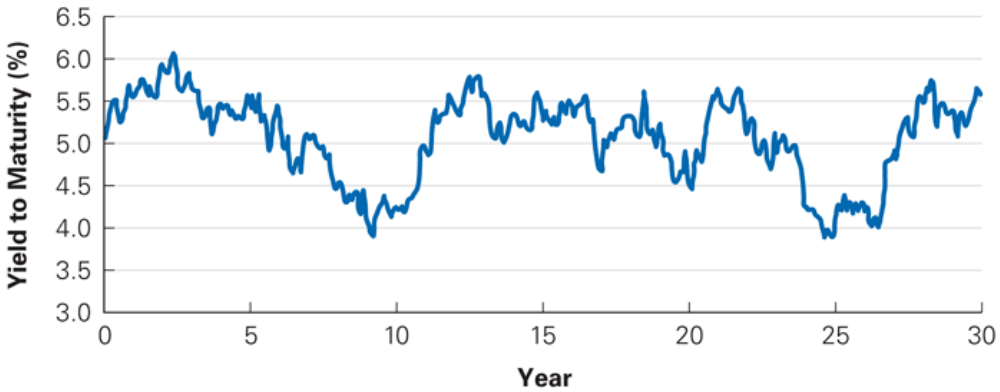
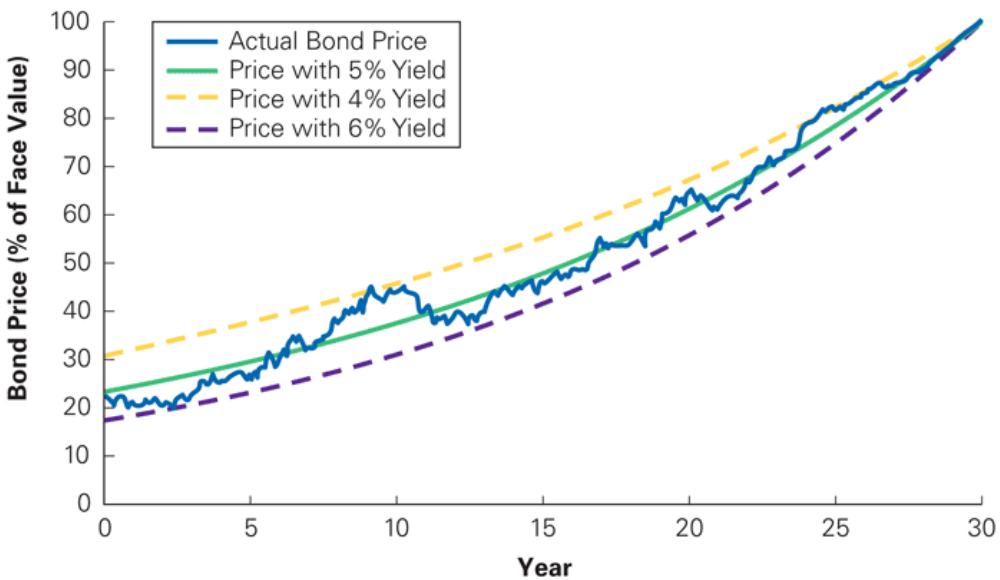
Interest rate changes

- There is an inverse relationship between interest rates and bond prices.
  - As interest rates and bond yields rise, bond prices fall.
  - As interest rates and bond yields fall, bond prices rise.

Interest rate changes





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Interest rate changes

- The sensitivity of a bond's price to changes in interest rates is measured by the bond's (modified) **duration** (See chapter 30.4)

$$-\frac{\frac{\Delta \text{Value}}{\text{Value}}}{\Delta \text{Yield}} = -\frac{\Delta \text{Value}}{\Delta \text{Yield}} \cdot \frac{1}{\text{Value}}$$

  - Bonds with high durations are highly sensitive to interest rate changes.
  - Bonds with low durations are less sensitive to interest rate changes.

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The Yield Curve and Bond Arbitrage

- Using the Law of One Price and the yields of default-free zero-coupon bonds, one can determine the price and yield of any other default-free bond.
- The yield curve provides sufficient information to evaluate all such bonds.

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Replicating a Coupon Bond

- Replicating a three-year $1000 bond that pays 10% annual coupon using three zero-coupon bonds:

	0	1	2	3
Coupon bond:		$100	$100	$1100
1-year zero:		$100		
2-year zero:			$100	
3-year zero:				$1100
Zero-coupon Bond portfolio:		$100	$100	$1100

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Replicating a Coupon Bond

- Yields and Prices (per $100 Face Value) for Zero Coupon Bonds:

Maturity	1 year	2 years	3 years	4 years
YTM	3.50%	4.00%	4.50%	4.75%
Price	$96.62	$92.45	$87.63	$83.06

Zero-Coupon Bond	Face Value Required	Cost
1 year		
2 years		
3 years		

Valuing a Coupon Bond Using Zero-Coupon Yields

- The price of a coupon bond must equal the present value of its coupon payments and face value discounted at the competitive market interest rates.
  - Price of a Coupon Bond:

$$PV = PV(\text{Bond Cash Flows})$$

$$= \frac{CPN}{1 + YTM_1} + \frac{CPN}{(1 + YTM_2)^2} + \cdots + \frac{CPN + FV}{(1 + YTM_n)^n}$$

$$P = \frac{100}{1.035} + \frac{100}{1.04^2} + \frac{100 + 1000}{1.045^3} = \$1153$$

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Corporate Bonds

- **Corporate Bonds:** Issued by corporations
- **Credit Risk:** Risk of default
- **Corporate Bond Yields**: Investors pay less for bonds with credit risk than they would for an otherwise identical default-free bond.
- The yield of bonds with credit risk will be higher than that of otherwise identical default-free bonds.

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Corporate Bond Yields

- No Default: Consider a 1-year, zero coupon Treasury Bill with a YTM of 4%. What is the price?

$$P = \frac{1000}{1 + YTM_1} = \frac{1000}{1.04} = \$961.54$$

- Certain Default: Suppose now bond issuer will pay 90% of the obligation. What is the price?

$$P = \frac{900}{1 + YTM_1} = \frac{900}{1.04} = \$865.38$$

- Certain Default: When computing the yield to maturity for a bond with certain default, the promised rather than the actual cash flows are used.

$$YTM = \frac{FV}{P} - 1 = \frac{1000}{865.38} - 1 = 15.56\%$$

Corporate Bonds

- **Certain Default:** The yield to maturity of a certain default bond is not equal to the expected return of investing in the bond. The yield to maturity will always be higher than the expected return of investing in the bond.
- **Risk of Default**
  - Consider a one-year, $1000, zero-coupon bond issued. Assume that the bond payoffs are uncertain. There is a 50% chance that the bond will repay its face value in full and a 50% chance that the bond will default and you will receive $900. Thus, you would expect to receive $950. Because of uncertainty, let the discount rate be 5.1 %. What is the price and YTM of the bond?

$$YTM = \frac{FV}{P} - 1 = \frac{1000}{903.90} - 1 = .1063$$

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Bonds Ratings

- Investment Grade Bonds (Top 4) versus Speculative Bonds (also known as Junk Bonds or High-Yield Bonds)

Rating*	Description (Moody's)
Investment Grade Debt	
Aaa/AAA	Judged to be of the best quality. They carry the smallest degree of investment risk and are generally referred to as "gilt edged." Interest payments are protected by a large or an exceptionally stable margin and principal is secure. While the various protective elements are likely to change, such changes as can be visualized are most unlikely to impair the fundamentally strong position of such issues.
Aa/AA	Judged to be of high quality by all standards. Together with the Aaa group, they constitute what are generally known as high-grade bonds. They are rated lower than the best bonds because margins of protection may not be as large as in Aaa securities or fluctuation of protective elements may be of greater amplitude or there may be other elements present that make the long-term risk appear somewhat larger than the Aaa securities.
A/A	Possess many favorable investment attributes and are considered as upper-medium-grade obligations. Factors giving security to principal and interest are considered adequate, but elements may be present that suggest a susceptibility to impairment some time in the future.
Baa/BBB	Are considered as medium-grade obligations (i.e., they are neither highly protected nor poorly secured). Interest payments and principal security appear adequate for the present but certain protective elements may be lacking or may be characteristically unreliable over any great length of time. Such bonds lack outstanding investment characteristics and, in fact, have speculative characteristics as well.

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Bonds Ratings

- Investment Grade Bonds versus Speculative Bonds (also known as Junk Bonds or High-Yield Bonds)

Speculative Bonds

Ba/BB	Judged to have speculative elements; their future cannot be considered as well assured. Often the protection of interest and principal payments may be very moderate, and thereby not well safeguarded during both good and bad times over the future. Uncertainty of position characterizes bonds in this class.
B/B	Generally lack characteristics of the desirable investment. Assurance of interest and principal payments of maintenance of other terms of the contract over any long period of time may be small.
Caa/CCC	Are of poor standing. Such issues may be in default or there may be present elements of danger with respect to principal or interest.
Ca/CC	Are speculative in a high degree. Such issues are often in default or have other marked shortcomings.
C/C, D	Lowest-rated class of bonds, and issues so rated can be regarded as having extremely poor prospects of ever attaining any real investment standing.

*Ratings: Moody's/Standard & Poor's

Source: www.moodys.com

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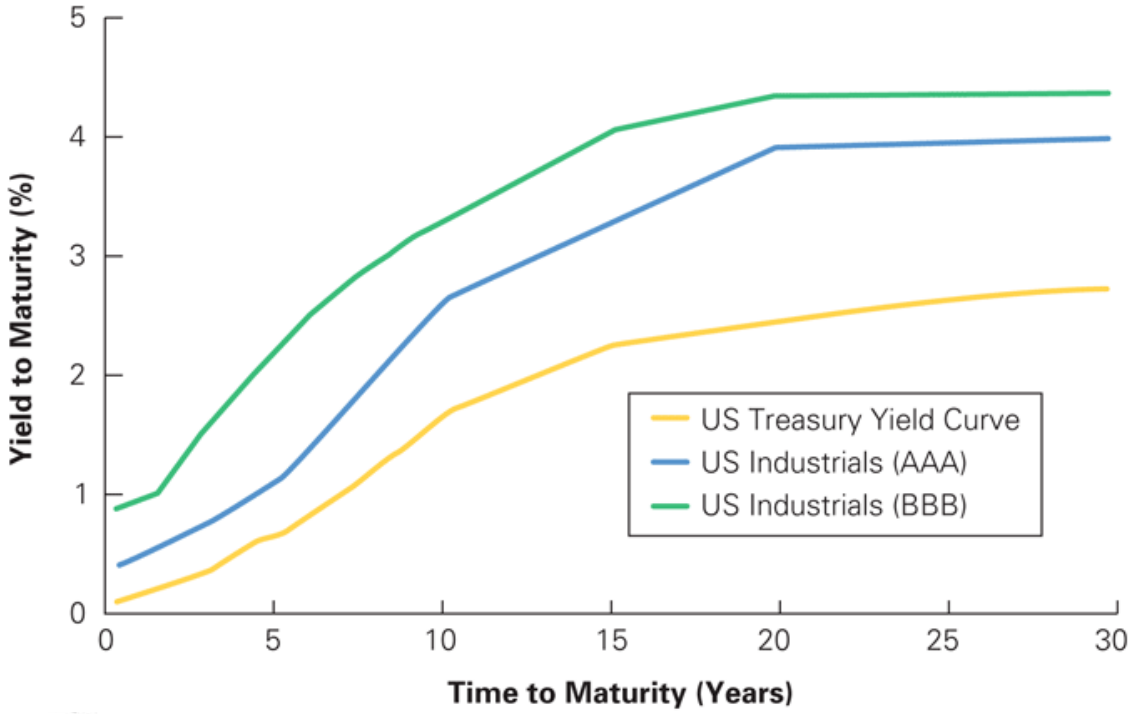
Corporate Yield Curves

- There is a **Default Spread** (also known as **Credit Spread**): The difference between the yield on corporate bonds and Treasury yields.



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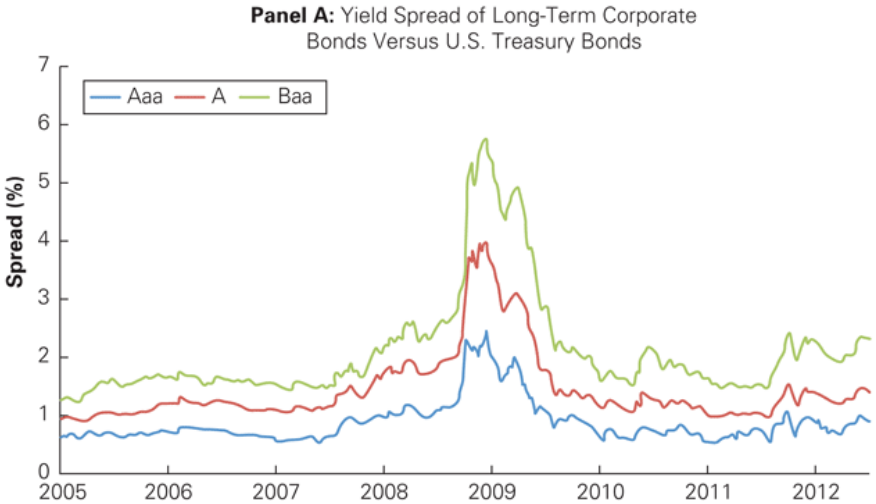
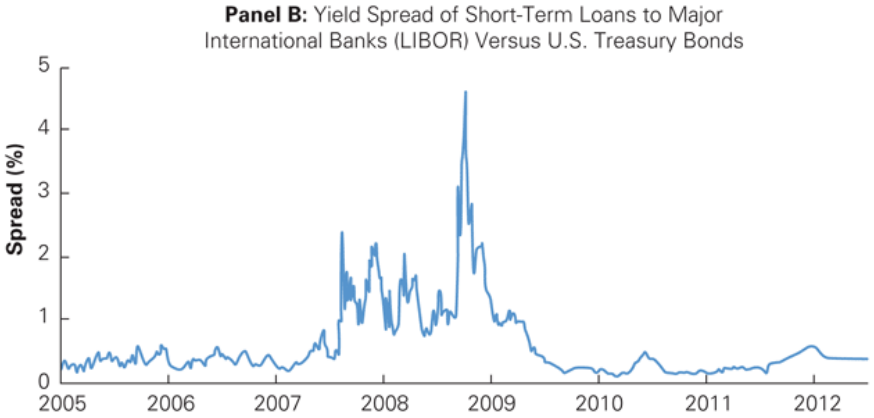
Yield Spreads and Financial Crisis





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Sovereign Bonds

- Bonds issued by national governments
- U.S. Treasury securities are generally considered to be default free
- All sovereign bonds are not default free
  - e.g. Greece defaulted on its outstanding debt in 2012
- Importance of inflation expectations
  - Potential to “inflate away” the debt
- European sovereign debt, the EMU, and the ECB

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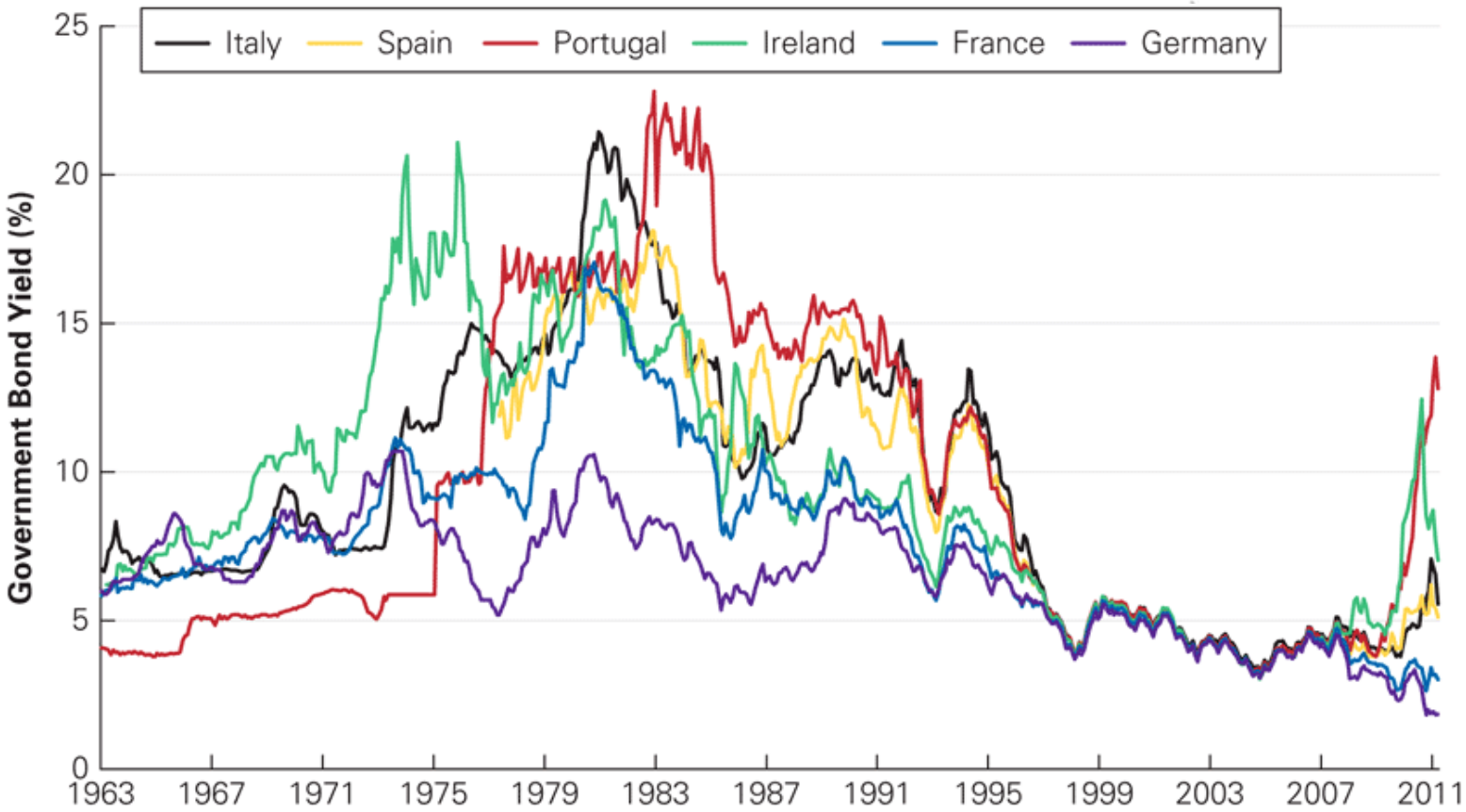
European Government Bond Yields, 1963–2011



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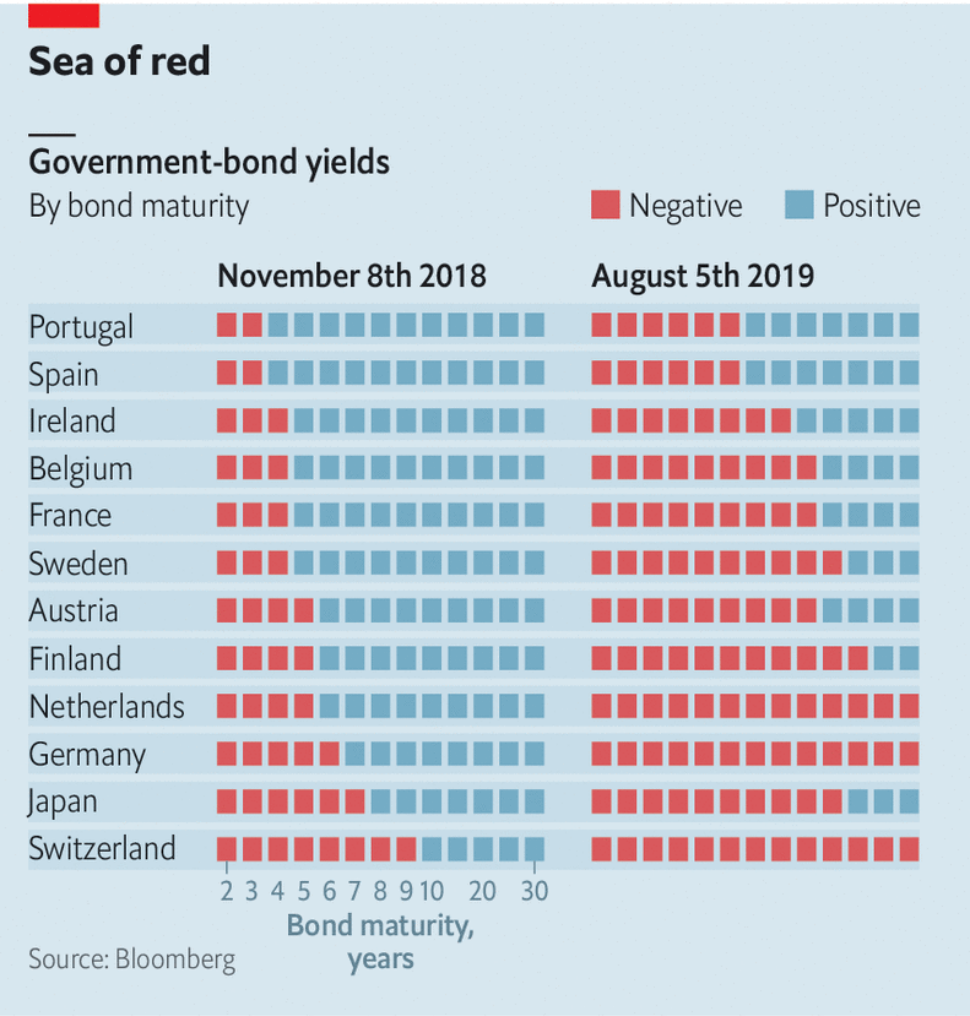
Can yields be negative?



The Economist

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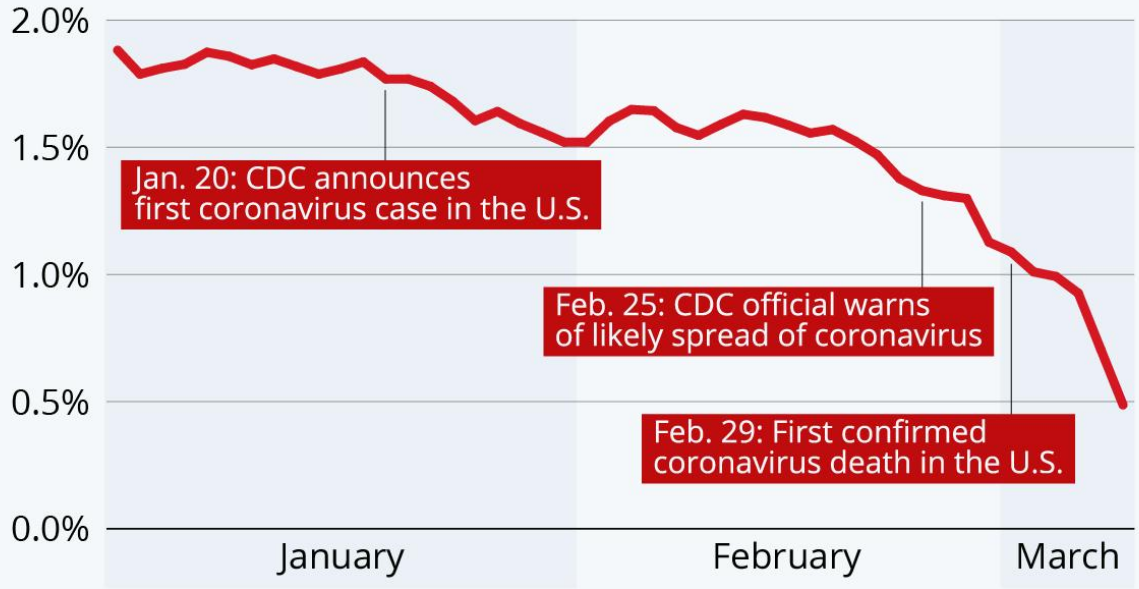
10-Year Treasury Yield Plummets to Record Low

Yield of 10-year U.S. Treasury bonds in 2020



As of March 9, 2020, 10:00 AM EDT
Source: Yahoo! Finance

COVID-19 Impact

statista

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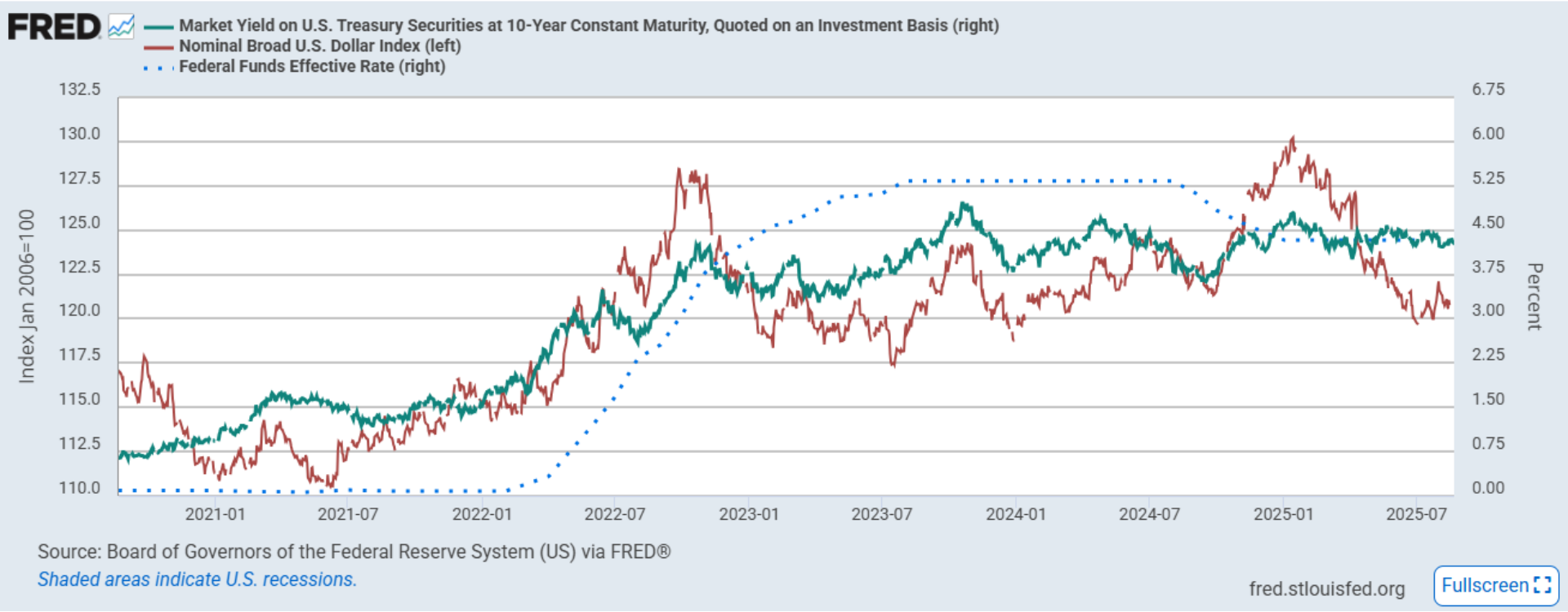
What is happening now?



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Appendix: Forward interest rate

- A forward interest rate (or forward rate) is an interest rate that we can guarantee today for a loan or investment that will occur in the future.
  - Here we only consider interest rate forward contracts for one-year investments: so the forward rate for year 5 means the rate available today on a one-year investment that begins four years from today.
  - We apply the Law of one price.
- Example: Suppose the one-year, zero-coupon yield is 5.5% and the two-year, zero-coupon yield is 7.0%. What is the one-year interest rate forward contract for year 2?

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Fundamentals of Capital Budgeting

- Chapter 8 learning objectives
  - You can **identify relevant cash flows** for a capital budgeting problem.
  - You can explain why **opportunity costs** must be included in cash flows, while **sunk costs and interest expense** must not.
  - You can calculate **taxes** that must be paid, including tax loss carryforwards and carrybacks.
  - You can calculate **free cash flows** for a given project.
  - You can illustrate the impact of **depreciation** on cash flows.
  - You know how to use **breakeven analysis, sensitivity analysis, or scenario analysis** to evaluate project risk.

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Terminology

- **Capital Budget**: Lists the investments that a company plans to undertake.
- **Capital Budgeting**: Process used to analyze alternate investments and decide which ones to accept.
- **Incremental Earnings**: The amount by which the firm's earnings are expected to change as a result of the investment decision.

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Forecasting Earnings

- Revenue Estimates for 4 year lifetime
  - Sales = 100,000 units/year
  - Per Unit Price = $260
- Cost/Expenses Estimates
  - Up-Front R&D = $15,000,000
  - Up-Front New Equipment = $7,500,000 with expected life of 5 years
  - New project will be housed in existing lab
  - Marketing and support costs= $2,800,000
  - Per Unit Cost = $110

Note: Cisco has already conducted a $ 300 000 feasibility study

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Forecasting Earnings

- Capital Expenditures and Depreciation
  - The $7.5 million in new equipment is a cash expense, but it is not directly listed as a cost when calculating earnings. Instead, the firm deducts a fraction of the acquisition cost of these items each year as depreciation.
  - *Straight Line Depreciation*: The asset's costs are divided equally over its life time. Annual Depreciation = $7.5 million ÷ 5 years = $1.5 million/year
- Interest Expenses
  - In capital budgeting decisions, interest expense is typically **not included**.
  - The rationale is that the project should be judged on its own, not on how it will be financed.

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Forecasting Earnings

- Taxes
  - Marginal Corporate Tax Rate: 40 % (the tax rate on the marginal or incremental dollar of pre-tax income)
  - Note: A negative tax is equal to a tax credit.
- Incremental Earnings Forecast

	Year	0	1	2	3	4	5
Incremental Earnings Forecast ($000s)							
1	Sales	—	26,000	26,000	26,000	26,000	—
2	Cost of Goods Sold	—	(11,000)	(11,000)	(11,000)	(11,000)	—
3	**Gross Profit**	—	15,000	15,000	15,000	15,000	—
4	Selling, General, and Administrative	—	(2,800)	(2,800)	(2,800)	(2,800)	—
5	Research and Development	(15,000)	—	—	—	—	
6	Depreciation	—	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7	**EBIT**	(15,000)	10,700	10,700	10,700	10,700	(1,500)
8	Income Tax at 40%	6,000	(4,280)	(4,280)	(4,280)	(4,280)	600
9	**Unlevered Net Income**	**(9,000)**	**6,420**	**6,420**	**6,420**	**6,420**	**(900)**

Incremental Earnings

- Indirect Effects on Incremental Earnings: Opportunity Cost
  - The value that a resource could have provided in its best alternative use.
  - Even though the equipment will be housed in an existing lab, the opportunity cost of not using the space in an alternative way (e.g., renting it out) must be considered: 200 000.
- Indirect Effects on Incremental Earnings: Project Externalities
  - Indirect effects of the project that may affect the profits of other business activities of the firm. Cannibalization is when sales of a new product displaces sales of an existing product.
  - Let sales of existing products drop by 2 500 000.
  - Accordingly, costs of goods sold for existing product can be saved: 1 500 000.

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Incremental Earnings Forecast Adjusted

	Year	0	1	2	3	4	5
Incremental Earnings Forecast ($000s)							
1	Sales	—	23,500	23,500	23,500	23,500	—
2	Cost of Goods Sold	—	(9,500)	(9,500)	(9,500)	(9,500)	—
3	**Gross Profit**	—	14,000	14,000	14,000	14,000	—
4	Selling, General, and Administrative	—	(3,000)	(3,000)	(3,000)	(3,000)	—
5	Research and Development	(15,000)	—	—	—	—	—
6	Depreciation	—	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7	**EBIT**	(15,000)	9,500	9,500	9,500	9,500	(1,500)
8	Income Tax at 40%	6,000	(3,800)	(3,800)	(3,800)	(3,800)	600
9	**Unlevered Net Income**	**(9,000)**	**5,700**	**5,700**	**5,700**	**5,700**	**(900)**

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Sunk Costs and Fixed Overhead Expenses

- **Sunk costs** or **irreversible** costs are costs that have been or will be paid regardless of the decision whether or not the investment is undertaken.
- Sunk costs should not be included in the incremental earnings analysis.
- For example money that has already been spent on R&D is a sunk cost and therefore irrelevant.
- **Fixed Overhead Expenses**: Typically, overhead costs are fixed and not incremental to the project and should not be included in the calculation of incremental earnings.

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Real-World Complexities

- Typically, sales will change from year to year.
- The selling price will vary over time.
- The cost per unit will change over time.
- Furthermore, all data are subject to uncertainty.

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Determining Free Cash Flow and NPV

- The incremental effect of a project on a firm's available cash is its free cash flow.
- **Capital Expenditures** are the actual cash outflows when an asset is purchased.
- These cash outflows are **included** in calculating free cash flow.
- **Depreciation** is a non-cash expense. The free cash flow estimate is **adjusted** for this **non-cash expense**.
- "Tax Rate times Depreciation" is called the **depreciation tax shield**.

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Net Working Capital (NWC)

- Most projects will require an investment in net working capital.
- **Net working capital** (NWC) is the difference between a company's current assets and current liabilities = Cash+Inventory+Trade credit
- **Trade credit** is the **difference** between **receivables** and **payables**.
- The increase in

$$\Delta NWC_t = NWC_t - NWC_{t-1}$$

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Calculating Free Cash Flow

$$\text{Free Cash Flow} = \overbrace{(\text{Revenues} - \text{Costs} - \text{Depreciation}) \times (1 - \tau_c)}^{\text{Unlevered Net Income}} + \text{Depreciation} - \text{CapEx} - \Delta NWC$$

$$\text{Free Cash Flow} = (\text{Revenues} - \text{Costs}) \times (1 - \tau_c) - \text{CapEx} - \Delta NWC + \tau_c \times \text{Depreciation}$$

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Determining Free Cash Flow

	Year	0	1	2	3	4	5
	Incremental Earnings Forecast ($000s)						
1	Sales	—	23,500	23,500	23,500	23,500	—
2	Cost of Goods Sold	—	(9,500)	(9,500)	(9,500)	(9,500)	—
3	**Gross Profit**	—	14,000	14,000	14,000	14,000	—
4	Selling, General, and Administrative	—	(3,000)	(3,000)	(3,000)	(3,000)	—
5	Research and Development	(15,000)	—	—	—	—	—
6	Depreciation	—	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
7	**EBIT**	(15,000)	9,500	9,500	9,500	9,500	(1,500)
8	Income Tax at 40%	6,000	(3,800)	(3,800)	(3,800)	(3,800)	600
9	**Unlevered Net Income**	**(9,000)**	**5,700**	**5,700**	**5,700**	**5,700**	**(900)**
	Free Cash Flow ($000s)						
10	Plus: Depreciation	—	1,500	1,500	1,500	1,500	1,500
11	Less: Capital Expenditures	(7,500)	—	—	—	—	—
12	Less: Increases in NWC	—	(2,100)	—	—	—	2,100
13	**Free Cash Flow**	**(16,500)**	**5,100**	**7,200**	**7,200**	**7,200**	**2,700**

Calculating the NPV

$$PV(FCF_t) = \frac{FCF_t}{(1+r)^t} = FCF_t \times \underbrace{\frac{1}{(1+r)^t}}_{t\ =\ \text{year discount factor}}$$

- Let the discount rate (required return) be 12 %.

	Year	0	1	2	3	4	5
Net Present Value ($000s)							
1 **Free Cash Flow**		(16,500)	5,100	7,200	7,200	7,200	2,700
2 Project Cost of Capital	12%						
3 Discount Factor		1.000	0.893	0.797	0.712	0.636	0.567
4 **PV of Free Cash Flow**		(16,500)	4,554	5,740	5,125	4,576	1,532
5 **NPV**		**5,027**					

- NPV = -16,500 + 4554 + 5740 + 5125 + 4576 + 1532

  = 5027

Choosing Among Alternatives

- Chapter 8.3 discusses how one should choose among different business alternatives (launching the project, not launching the project, evaluating manufacturing alternatives etc.)
- When choosing between mutually exclusive investment opportunities, pick the opportunity with the highest NPV.
- When analyzing alternatives, we only need to include those components of free cash flow that differ among the alternatives.

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Further Adjustments to Free Cash Flow (examples given in the textbook)

- Depreciation expenses affect free cash flow only through the tax shield. The firm should generally use the most accelerated depreciation schedule allowed
- Free cash flow should also include the **after tax liquidation or salvage value** of the asset.
- When an asset is sold, tax is due to the difference between the sales price and the asset's book value net of depreciation (**capital gain or loss**)
- **Terminal** or **continuation value**s should reflect the present value of the project's future cash flows **beyond the forecast horizon**

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Analyzing the Project
(examples given in the textbook)

- **Break even analysis** computes the level of a parameter that makes the project's NPV equal zero
- **Sensitivity analysis** shows how NPV changes the value of one (or two) parameter change over a range (keeping the other parameters fixed)
- **Scenario analysis** considers the effect of changing multiple parameter simultaneously

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